

- a. Write a Java program to create an abstract class `CreditCard` with basic card details, a method `displayDetails()` and abstract method `calculateBill()`. Create subclasses: `SilverCard`, `GoldCard` and `PlatinumCard`. When the user chooses the card and provides purchase amount corresponding card details, features and final bill amount should be displayed.
- `SilverCard` has following features
 - Maximum credit limit – 2,00,000
 - No Discount on purchase
 - `GoldCard` has following features
 - Maximum credit limit – 5,00,000
 - 5% Discount on purchase
 - `PlatinumCard` has following features
 - Maximum credit limit – 10,00,000
 - 10% Discount on purchase
- b. Write a Java program to create an interface `Shape` with methods `calculateArea()` and `calculateVolume()`. Create a constant to store the value of π . Create three classes `Sphere`, `Cylinder` and `Cone` that implements the `Shape` interface.